

human

Show me the top 10 largest bulding by area in Exeter. Plot the buildings correctly

host_agent

Tool call: generate_metadata_for_all_artifacts with args: {} Tool response: {}

Tool call: send_message with args: {'target': 'planning_agent', 'task_description': 'User wants to find and plot the top 10 largest buildings by area in Exeter. Provide the general steps to solve this query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, specifically the largest ones by area. The condition is to find the top 10 largest buildings.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter as many entries as possible", "Sort buildings by area", "Select the top 10 largest buildings", "Plot the top 10 largest buildings"]

host_agent

Tool send_message returned result: The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, specifically the largest ones by area. The condition is to find the top 10 largest buildings.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter as many entries as possible", "Sort buildings by area", "Select the top 10 largest buildings", "Plot the top 10 largest buildings"]

Tool call: send_message with args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Tool call: call_os_ngd with args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} Tool response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of

records fetched : 98.

host_agent

Tool call: send_message with args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Tool call: generate_metadata_for_artifacts with args: {'artifact_names': ['exeter_area_polygon']} Tool response: (metadata generated)

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the name or description indicates 'Exeter' and is a city or settlement
    mask = (
        (gdf['name1_text'].str.lower() == 'exeter') |
        (gdf['description'].str.lower().str.contains('city')) |
        (gdf['descriptiongroup'].str.lower().str.contains('settlement'))
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'Polygon(s) from exeter_area_polygon filtered to represent the city of Exeter.'
    return [summary, artifact_name, artifact_description, filtered]

filter_exeter_area_polygon(data)
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the name or description indicates 'Exeter' and is a city or settlement
    mask = (
        (gdf['name1_text'].str.lower() == 'exeter') |
        (gdf['description'].str.lower().str.contains('city')) |
        (gdf['descriptiongroup'].str.lower().str.contains('settlement'))
```

```
)
filtered = gdf[mask]
summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
artifact_name = 'filtered_exeter_area_polygon'
artifact_description = 'Polygon(s) from exeter_area_polygon filtered to represent the city of Exeter.'
return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result: ['Filtered to 34 polygon(s) for Exeter.',]

Filtered to 34 polygon(s) for Exeter. An artifact has been generated with the name 'filtered_exeter_area_polygon' and contains polygons from 'exeter_area_polygon' filtered to represent the city of Exeter.

named_area_agent

The specific polygon(s) for Exeter have been filtered and saved as 'filtered_exeter_area_polygon'.

host_agent

Tool call: send_message with args: {'target': 'buildings', 'task_description': "Search for all buildings within the 'filtered_exeter_area_polygon' artifact."}

buildings_agent

Tool call: call_os_ngd with args: {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_buildings'} Tool response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for all buildings within the 'filtered_exeter_area_polygon' artifact. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for all buildings within the 'filtered_exeter_area_polygon' artifact. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for all buildings within the 'filtered_exeter_area_polygon' artifact. using filters None."], counts of records fetched are : [72895, 72364, 89825].

I found the following artifacts for buildings within the 'filtered_exeter_area_polygon':

1. **buildings_exeter_buildings**: A geopandas dataframe containing building data. (Records: 72,895)
2. **buildingline_exeter_buildings**: A geopandas dataframe containing building line data. (Records: 72,364)
3. **buildingpart_exeter_buildings**: A geopandas dataframe containing building part data. (Records: 89,825)

These artifacts contain all the building data within the specified area. If you need further analysis or filtering, please let me know!

host_agent

Tool call: send_message with args: {'target': 'plotting_agent', 'task_description': "From the 'buildings_exeter_buildings' artifact, select and plot the top 10 largest buildings by area within Exeter. Provide a summary and the map artifact."}

plotting_agent

Tool call: generate_metadata_for_artifacts with args: {'artifact_names': ['buildings_exeter_buildings']} Tool response: (metadata generated)

Tool Code Executor

```
def plot_top_10_largest_buildings_by_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    gdf = data[0].copy()
    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_string() != 'EPSG:4326':
        gdf = gdf.to_crs(epsg=4326)

    # Sort by area and select top 10
    top10 = gdf.sort_values('geometry_area_m2', ascending=False).head(10)

    # Center map on the centroid of the top 10
    centroid = top10.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=15)

    # Add all top 10 buildings to the map
    for _, row in top10.iterrows():
        folium.GeoJson(
            mapping(row['geometry']),
            name=f"{row.get('description', 'Building')} (Area: {row['geometry_area_m2']:.0f} m²)",
            tooltip=f"Description: {row.get('description', 'N/A')}<br>Area: {row['geometry_area_m2']:.2f} m²"
        ).add_to(m)

    m.save('top10_largest_buildings_exeter.html')
    summary = f"Top 10 largest buildings by area in Exeter.\n" + \
        top10[['osid', 'description', 'geometry_area_m2']].head(5).to_string(index=False) + \
        f"\nTotal buildings plotted: {len(top10)}. Area range: {top10['geometry_area_m2'].min():.2f} - {top10['geometry_area_m2'].max():.2f} m²."
    return [summary, 'top10_largest_buildings_map', 'Folium map showing the top 10 largest buildings by area in Exeter, with tooltips for description and area.']
```

Tool code_executor returned result: ['Top 10 largest buildings by area in Exeter.\n osid description geometry_area_m2\n9ca476de-7c93-4335-b58b-83d9752e8058 Hospital 42567.469\n1aa9c75c-ada0-40b3-8e87-9a201d2be7b6 Commercial Building 19994.457\nf0383ff6-f00b-425b-9bae-f9c426461a2e Mixed Use Building 16769.611\n8fec21e8-feda-48a3-8d1e-b6e2c981e64a Central Government Services 15440.779\n092419db-9512-40f1-9fa8-37c70e3a528a Library 12883.372\nTotal buildings plotted: 10. Area range: 9644.11 - 42567.47 m².',]

['Top 10 largest buildings by area in Exeter.\n osid description geometry_area_m2\n9ca476de-7c93-4335-b58b-83d9752e8058 Hospital 42567.469\n1aa9c75c-ada0-40b3-8e87-9a201d2be7b6 Commercial Building 19994.457\nf0383ff6-f00b-425b-9bae-f9c426461a2e Mixed Use Building 16769.611\n8fec21e8-feda-48a3-8d1e-b6e2c981e64a Central Government Services 15440.779\n092419db-9512-40f1-9fa8-37c70e3a528a Library 12883.372\nTotal buildings plotted: 10. Area range: 9644.11 - 42567.47 m².', 'top10_largest_buildings_map', 'Folium map showing the top 10 largest buildings by area in Exeter, with tooltips for description and area.', 'top10_largest_buildings_exeter.html']

host_agent

The top 10 largest buildings by area in Exeter have been identified and plotted. Here are some of the largest buildings:

1. Hospital – 42,567 m²
2. Commercial Building – 19,994 m²
3. Mixed Use Building – 16,769 m²
4. Central Government Services – 15,440 m²
5. Library – 12,883 m²

A total of 10 buildings were plotted, with areas ranging from 9,644 m² to 42,567 m².

You can view the map in the artifact: "top10_largest_buildings_map", which shows the locations and details of these buildings in Exeter. If you need the map file or further details, let me know!